13

Forecasting Performance

This chapter focuses on the *mechanics* of forecasting—specifically, how to develop an integrated set of financial forecasts. We'll explore how to build a well-structured spreadsheet model: one that separates raw inputs from computations, flows from one worksheet to the next, and is flexible enough to handle multiple scenarios. Then we'll discuss the process of forecasting.

To arrive at future cash flow, we forecast the income statement, balance sheet, and statement of changes in equity. The forecast financial statements provide the information necessary to compute net operating profit after taxes (NOPAT), invested capital, return on invested capital (ROIC), and, ultimately, free cash flow (FCF).

While you are building a forecast, it is easy to become engrossed in the details of individual line items. But we stress the importance of placing your aggregate results in the proper context. You can do much more to improve your valuation through a careful analysis of whether your forecast of future ROIC is consistent with the company's ability to generate value than you can by precisely (but perhaps inaccurately) forecasting an immaterial line item ten years out.

DETERMINE THE FORECAST'S LENGTH AND DETAIL

Before you begin forecasting individual line items on the financial statements, decide how many years to forecast and how detailed your forecast should be. The typical solution, described in Chapter 10, is to develop an explicit year-by-year forecast for a set period and then to value the remaining years by using a perpetuity formula, such as the key value driver formula introduced in Chapter 3. Whatever perpetuity formula you choose, all the continuing-value approaches assume steady-state performance. Thus, the explicit forecast

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period must be long enough for the company to reach a steady state, defined by the following characteristics:

- The company grows at a constant rate by reinvesting a constant proportion of its operating profits into the business each year.
- The company earns a constant rate of return on both existing capital and new capital invested.

As a result, free cash flow for a steady-state company will grow at a constant rate and can be valued using a growth perpetuity. The explicit forecast period should be long enough that the company's growth rate is less than or equal to that of the economy. Higher growth rates would eventually make companies unrealistically large relative to the aggregate economy.

In general, we recommend using an explicit forecast period of 10 to 15 years—perhaps longer for cyclical companies or those experiencing very rapid growth. Using a short explicit forecast period, such as five years, typically results in a significant undervaluation of a company or requires heroic long-term growth assumptions in the continuing value. Even so, a long forecast period raises its own issues—namely, the difficulty of forecasting individual line items 10 to 15 years into the future.

To simplify the model and avoid the error of false precision, we often split the explicit forecast into two periods:

1. A detailed five-year to seven-year forecast, which develops complete balance sheets and income statements with as many links as possible to real variables such as unit volumes and cost per unit
2. A simplified forecast for the remaining years, focusing on a few important variables, such as revenue growth, margins, and capital turnover

Using a simplified intermediate forecast forces you to focus on the business's long-term economics, rather than become engrossed in too much detail.

COMPONENTS OF A GOOD MODEL

If you combine 15 years of financial forecasts with 10 years of historical analysis, even the simplest valuation spreadsheet becomes complex. Therefore, you should carefully design and structure your model before starting to forecast. In Exhibit 13.1, we structure a valuation model with seven distinct worksheets:

1. *Raw historical data.* Collect raw data from the company's financial statements, footnotes, and external reports in one place.[1] By keeping the data

[1] For large, established companies, the amount of collected data can be substantial. To analyze Costco in Chapter 11 and Appendix H, we created separate worksheets for the company's financial statements, statutory tax table, and note on deferred taxes.

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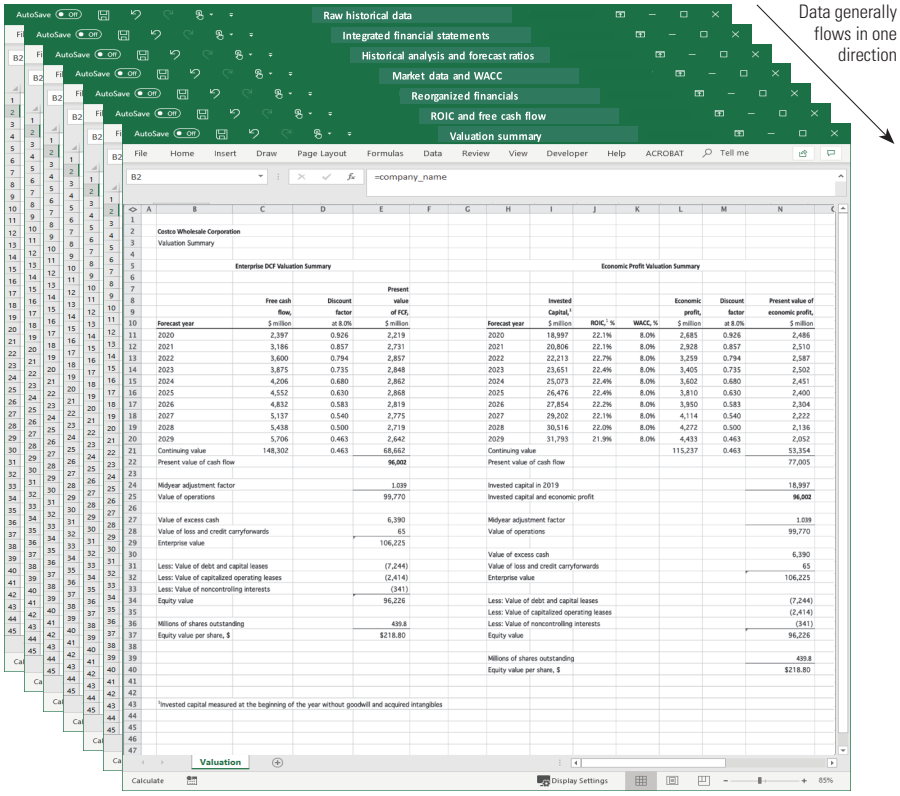
EXHIBIT 13.1 **Sample Workbook**



Costco Wholesale Corporation
Valuation Summary

Enterprise DCF Valuation Summary

Forecast year	Free cash flow, $ million	Discount factor at 8.0%	Present value of FCF, $ million
2020	2,397	0.926	2,219
2021	3,186	0.857	2,731
2022	3,600	0.794	2,857
2023	3,875	0.735	2,848
2024	4,206	0.680	2,862
2025	4,552	0.630	2,868
2026	4,832	0.583	2,819
2027	5,137	0.540	2,775
2028	5,438	0.500	2,719
2029	5,706	0.463	2,642
Continuing value	148,302	0.463	68,662
Present value of cash flow			96,002
Midyear adjustment factor			1.039
Value of operations			99,770
Value of excess cash			6,390
Value of loss and credit carryforwards			65
Enterprise value			106,225
Less: Value of debt and capital leases			(7,244)
Less: Value of capitalized operating leases			(2,414)
Less: Value of noncontrolling interests			(341)
Equity value			96,226
Millions of shares outstanding			439.8
Equity value per share, $			$218.80

Economic Profit Valuation Summary

Forecast year	Invested Capital,[1] $ million	ROIC,[1] %	WACC, %	Economic profit, $ million	Discount factor at 8.0%	Present value of economic profit, $ million
2020	18,997	22.1%	8.0%	2,685	0.926	2,486
2021	20,806	22.1%	8.0%	2,928	0.857	2,510
2022	22,213	22.7%	8.0%	3,259	0.794	2,587
2023	23,651	22.4%	8.0%	3,405	0.735	2,502
2024	25,073	22.4%	8.0%	3,602	0.680	2,451
2025	26,476	22.4%	8.0%	3,810	0.630	2,400
2026	27,854	22.2%	8.0%	3,950	0.583	2,304
2027	29,202	22.1%	8.0%	4,114	0.540	2,222
2028	30,516	22.0%	8.0%	4,272	0.500	2,136
2029	31,793	21.9%	8.0%	4,433	0.463	2,052
Continuing value				115,237	0.463	53,354
Present value of cash flow						77,005
Invested capital in 2019						18,997
Invested capital and economic profit						96,002
Midyear adjustment factor						1.039
Value of operations						99,770
Value of excess cash						6,390
Value of loss and credit carryforwards						65
Enterprise value						106,225
Less: Value of debt and capital leases						(7,244)
Less: Value of capitalized operating leases						(2,414)
Less: Value of noncontrolling interests						(341)
Equity value						96,226
Millions of shares outstanding						439.8
Equity value per share, $						$218.80

[1]Invested capital measured at the beginning of the year without goodwill and acquired intangibles

together, you can verify information as needed and update data year by year. Report the raw data in their original form.

2. *Integrated financial statements.* Using figures from the raw-data worksheet, create a set of historical financials that find the right level of detail. As a general rule, operating and nonoperating items should not be aggregated within the same line item. The income statement should be linked with the balance sheet through retained earnings. This worksheet will contain historical and forecast financial statements.
3. *Historical analysis and forecast ratios.* For each line item in the financial statements, build historical ratios, as well as forecasts of future ratios. These ratios will generate the forecast financial statements contained on the previous sheet.
4. *Market data and weighted average cost of capital (WACC).* Collect all financial market data on one worksheet. This worksheet will contain estimates of beta, the cost of equity, the cost of debt, and the weighted average cost of capital, as well as historical market values and trading multiples for the company.
5. *Reorganized financial statements.* Once you have built a complete set of financial statements (both historical and forecast), reorganize the financial

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statements to calculate NOPAT, its reconciliation to net income, invested capital, and its reconciliation to total funds invested.

6. *ROIC and FCF.* Use the reorganized financials to build return on invested capital, economic profit, and free cash flow. Future free cash flow will be the basis of your enterprise valuation.
7. *Valuation summary.* Create a summary worksheet that sums discounted cash flows and converts the value of operations into equity value. The valuation summary includes the value of operations, value of nonoperating assets, value of nonequity claims, and resulting equity value.

Well-built valuation models have certain characteristics. First, original data and user input are collected in only a few places. For instance, limit original data and user input to just three worksheets: raw data (worksheet 1), forecasts (worksheet 3), and market data (worksheet 4). To provide additional clarity, denote raw data and user input in a different color from calculations. Second, whenever possible, a given worksheet should feed into the next worksheet. Formulas should not bounce from sheet to sheet without clear direction.[2] Raw data should feed into integrated financials, which in turn should feed into ROIC and FCF. Third, unless specified as data input, numbers should never be hard-coded into a formula. Hard-coded numbers are easily forgotten as the spreadsheet grows in complexity. Finally, use formulas that come built into the spreadsheet software sparingly, such as the net present value (NPV) formula. Built-in formulas can obscure the model's logic and make auditing results difficult.

MECHANICS OF FORECASTING

The enterprise discounted-cash-flow (DCF) valuation model relies on a forecast of free cash flow (FCF). However, as noted at the beginning of this chapter, FCF forecasts should be created indirectly by first forecasting the income statement, balance sheet, and statement of retained earnings. Compute forecasts of free cash flow in the same way as when analyzing historical performance. (A well-built spreadsheet will use the same formulas for historical and forecast ROIC and FCF without any modification.)

We break the forecasting process into six steps:

1. *Prepare and analyze historical financials.* Before forecasting future financials, you must build and analyze historical financials. A robust analysis will place your forecasts in the appropriate context.
2. *Build the revenue forecast.* Almost every line item will rely directly or indirectly on revenues. Estimate future revenues by using either a top-down

[2] Data should always flow in one direction and never loop back to create a circular reference. Circular references will prevent your spreadsheet from calculating results accurately.

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(market-based) or a bottom-up (customer-based) approach. Forecasts should be consistent with evidence on growth.

3. *Forecast the income statement.* Use the appropriate economic drivers to forecast operating expenses, depreciation, nonoperating income, interest expense, and reported taxes.
4. *Forecast the balance sheet: invested capital and nonoperating assets.* On the balance sheet, forecast operating working capital, net property, plant, and equipment, goodwill, and nonoperating assets.
5. *Reconcile the balance sheet with investor funds.* Complete the balance sheet by computing retained earnings and forecasting other equity accounts. Use excess cash and/or new debt to balance the balance sheet.
6. *Calculate ROIC and FCF.* Calculate ROIC on future financial statements to ensure your forecasts are consistent with economic principles, industry dynamics, and the company's ability to compete. To complete the forecast, calculate free cash flow as the basis for valuation. Future FCF should be calculated the same way as historical FCF.

Give extra emphasis to your revenue forecast. Almost every line item in the spreadsheet will be either directly or indirectly driven by revenues, so you should devote enough time to arrive at a good revenue forecast, especially for rapidly growing businesses.

Step 1: Prepare and Analyze Historical Financials

Before starting to build a forecast, input the company's historical financials into a spreadsheet. To do this, you can rely on data from a professional service, such as Bloomberg, Capital IQ, Compustat, or Thomson ONE, or you can use financial statements directly from the company's filings.

Professional services offer the benefit of standardized data (i.e., financial data formatted into a set number of categories). Since data items do not change across companies, a single spreadsheet can quickly analyze any company. However, using a standardized data set carries a cost. Many of the specified categories aggregate important items, hiding critical information. For instance, Compustat groups "advances to sales staff" (an operating asset) and "pension and other special funds" (a nonoperating asset) into a single category titled "other assets." Because of this, models based solely on preformatted data can lead to meaningful errors in the estimation of value drivers, and hence to poor valuations.

Alternatively, you can build a model using financials from the company's annual report. To use raw data, however, you must dig. Often, companies aggregate critical information to simplify their financial statements. Consider, for instance, the financial data for Honeywell presented in Exhibit 13.2. On Honeywell's reported balance sheet, the company consolidates many items into the account titled "accrued liabilities." In the notes that follow the company's

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EXHIBIT 13.2 **Honeywell: Current Liabilities in Balance Sheet**

$ million

Balance Sheet	2017	2018
Accounts payable	6,584	5,607
Commercial paper and other short-term borrowings	3,958	3,586
Current maturities of long-term debt	1,351	2,872
→ Accrued liabilities	6,968	6,859
Total current liabilities	18,861	18,924
Note 12: Accrued liabilities		
Customer advances and deferred income	2,198	2,403
Compensation, benefit, and other employee-related costs	1,420	1,469
Asbestos-related liabilities	350	245
Repositioning	508	566
Product warranties and performance guarantees	307	243
Environmental costs	226	175
Income taxes	134	166
Accrued interest	94	94
Other taxes	277	234
Insurance	199	170
Other (primary operating expenses)	1,255	1,094
→ Accrued liabilities	6,968	6,859

Source: Honeywell International annual report, 2018.

financial statements, note 12 details this line item. Some of the components (such as compensation, benefit, and other employee-related costs) are operating liabilities, and others are debt equivalents (such as environmental costs). Since the valuation of each of these items requires different treatment, the items must be separated on the expanded balance sheet.

We prefer to collect raw data on a separate worksheet. On the raw-data sheet, record financial data as originally reported, and never combine multiple data into a single cell. Once you have collected raw data from the reported financials and notes, use the data to build a set of expanded (or simplified) financial statements: the income statement, balance sheet, statement of equity, and statement of accumulated other comprehensive income. Although the statement of equity appears redundant, it will be critical for error checking during the forecasting process, because it connects the income statement to the balance sheet. If available, accumulated other comprehensive income will be necessary to complete the free cash flow statement.

As you build the integrated financials, you must decide whether to aggregate immaterial line items. Analyzing and forecasting too many line items can lead to confusion, introduce errors, and cause the model to become unwieldy. Returning to the Honeywell example presented in Exhibit 13.2, the income taxes payable account amounts to under 0.1 percent of Honeywell's revenues.[3] Therefore, you might simplify a valuation of Honeywell by combining income

[3] Contrast this to accrued compensation and employee benefit costs; that account is nearly 15 times as large as taxes payable. Given its size, accrued compensation and employee benefit costs should not be aggregated with other accrued liabilities.

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taxes payable with the "other" account. When aggregating, however, make sure never to combine operating and nonoperating accounts into a single category. If operating and nonoperating accounts are combined, you cannot calculate ROIC and FCF properly.

Step 2: Build the Revenue Forecast

To build a revenue forecast, you can use a *top-down* forecast, in which you estimate revenues by sizing the total market, determining market share, and forecasting prices. Alternatively, with the *bottom-up* approach, you can use the company's own forecasts of demand from existing customers, customer turnover, and the potential for new customers. When possible, use both methods to establish bounds for the forecast.

The top-down approach can be applied to any company. For companies in mature industries, the aggregate market grows slowly and is closely tied to economic growth and other long-term trends, such as changing consumer preferences. In these situations, you can rely on third-party forecasts of the aggregate market and focus your own efforts on forecasting market share by competitor.[4] To do this, you must determine which companies have the capabilities and resources to compete effectively and capture share. A good place to start, of course, is with historical financial analysis. But more importantly, make sure to address how the company is positioned for the future. Does it have the required products and services to capture share? Do other competitors have products and services that will displace the company's market position? A good forecast will address each of these issues.

Over the short term, top-down forecasts should build on the company's announced intentions and capabilities for growth. For instance, retailers like Costco have well-mapped plans for new store openings, which are their primary driver of revenue growth. Oil companies like BP have proven reserves and relatively fixed amounts of refining capacity. And pharmaceutical companies like Merck have a fixed set of drugs under patent and in clinical trials.

To value Costco in Appendix H, we relied on forecasts from the sell-side analyst community to project company revenue. Exhibit 13.3 presents one analyst's forecasts for Costco. The forecast is split into domestic revenues, international revenues, membership fees, and ancillary businesses. The ancillary-businesses segment includes gas stations and pharmacies. Using a geographically segmented forecast for Costco is important because revenue per square foot and square feet per store differ between domestic and international stores. Details such as these will vary depending on the company and industry. Some analysts will provide their corporate clients with forecasts of the number of transactions, average revenue per transaction, and other

[4] Examples of third-party forecasts include EvaluatePharma for drug-by-drug revenue forecasts, McCoy Power Reports for power generation equipment, and RBR for point-of-sale systems.

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EXHIBIT 13.3 **Costco: Sample Revenue Forecast[1]**

$ million

	Historical			Forecast					
	2017	2018	2019	2020	2021	2022	2023	2024	2025
U.S. revenues									
Revenue per square foot, $	1,007	1,054	1,100	1,144	1,172	1,202	1,226	1,250	1,275
× Square footage per store, thousands	147	147	147	147	148	148	148	148	148
× Number of stores	514	527	543	558	566	574	582	590	598
= U.S. revenues	76,087	81,652	87,803	93,838	98,176	102,112	105,603	109,150	112,843
International stores									
Revenue per square foot, $	904	958	968	997	1,027	1,058	1,089	1,122	1,156
× Square footage per store, thousands	142	142	144	144	144	144	144	144	144
× Number of stores	225	233	236	244	252	260	268	276	284
= International revenues	28,883	31,696	32,897	35,031	37,268	39,612	42,027	44,593	47,276
Membership fees									
Average fee per member	32	33	34	35	35	36	37	38	38
× Number of members, millions	90	94	99	102	104	108	110	114	116
= Membership fees	2,853	3,140	3,349	3,539	3,682	3,899	4,048	4,286	4,443
Ancillary businesses[2]	21,400	24,900	28,600	30,900	33,400	36,100	39,000	42,100	45,500
Total revenues	129,223	141,389	152,649	163,308	172,525	181,723	190,677	200,129	210,061

[1] For better comparability across companies, data are presented on a calendar basis. Costco's fiscal year-end is August 31.

[2] Ancillary businesses include gas stations, pharmacies, optical dispensing centers, food courts, and hearing-aid centers.

Source: Trefis, "Costco," November 2019.

revenue drivers. Taking a fine-grained look at a company's sources of growth will make clear what drives the company's valuation.

In new-product markets, the top-down approach is especially helpful but often requires more work than for established markets. For instance, consider the recent launch of June Life, a maker of web-enabled ovens. The company's smart oven is marketed as many appliances in one, including a toaster, dehydrator, and slow cooker. The accompanying smartphone app allows the user to control the oven remotely, check on remaining time, and even view the product cooking.

Given the lack of history for the company's products, how do you estimate the potential size and speed of penetration of this new product? You could start by sizing the more traditional products of Black & Decker and Cuisinart. Analyze whether the new smart ovens, given their greater functionality, will be adopted by even more users than traditional ovens—or perhaps by fewer, because of their higher price. Next, forecast how quickly web-enabled products might penetrate the market. To do this, look at the speed of migration for other electronics that have gone through a similar transition, such as the voice-only cell phone to the smartphone. Determine the characteristics that drive conversion in other markets; this helps you place your forecast in context. Next, assess the price point and resulting operating margin for the company's products. How many companies are developing the product, and

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how competitive will the market be? As you can see, there are more questions than answers. The key is structuring the analysis and applying historical evidence from comparable markets to help bound forecasts whenever possible.

Whereas a top-down approach starts with the aggregate market and predicts penetration rates, price changes, and market shares, a bottom-up approach relies on projections of customer demand. In some industries, a company's customers will have projected their own revenue forecasts and can give their suppliers a rough estimate of their own purchase projections. By aggregating across customers, you can determine short-term forecasts of revenues from the current customer base. Next, estimate the rate of customer turnover. If customer turnover is significant, you must eliminate a portion of estimated revenues. As a final step, project how many new customers the company will attract and how much revenue those customers will contribute. The resulting bottom-up forecast combines new customers with revenues from existing customers.[5]

Regardless of the method, forecasting revenues over long time periods is imprecise. Customer preferences, technologies, and corporate strategies change. These often-unpredictable changes can profoundly influence the winners and losers in the marketplace. Therefore, you must constantly reevaluate whether the current forecast is consistent with industry dynamics, competitive positioning, and the historical evidence on corporate growth. If you lack confidence in your revenue forecast, use multiple scenarios to model uncertainty. Doing this not only will bound the forecast, but also will help company management make better decisions. A discussion of scenario analysis can be found in Chapter 16.

Step 3: Forecast the Income Statement

With a revenue forecast in place, forecast individual line items related to the income statement. To forecast a line item, use a three-step process:

1. *Decide what economic relationships drive the line item*. For most line items, forecasts will be tied directly to revenues. Some line items will be economically tied to a specific asset or liability. For instance, interest income is usually generated by cash and marketable securities; if this is the case, forecasts of interest income should be tied to cash and marketable securities.
2. *Estimate the forecast ratio*. For each line item on the income statement, compute historical values for each ratio, followed by estimates for each of the forecast periods. To get the model working properly, initially set the forecast ratio equal to the previous year's value. Your forecasts are

[5] For more on company valuation using customer acquisition and retention statistics, see Daniel McCarthy, Peter Fader, and Bruce Hardie, "Valuing Subscription-Based Businesses Using Publicly Disclosed Customer Data," *Journal of Marketing* 81, no. 1 (2018): 17–35.

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EXHIBIT 13.4 **Partial Forecast of the Income Statement**

Forecast worksheet

%	2019	Forecast 2020
Revenue growth	20.0	20.0
Cost of goods sold/revenues	37.5	37.5
Selling and general expenses/revenues	18.8	
Depreciation$_t$/net PP&E$_{t-1}$[1]	9.5	

Step 1: Choose a forecast driver, and compute historic ratios.

Step 2: Estimate the forecast ratio.

Income statement

$ million	2019	Forecast 2020
Revenues	240.0	288.0
Cost of goods sold	(90.0)	(108.0)
Selling and general expenses	(45.0)	
Depreciation	(19.0)	
EBITA	86.0	
Interest expense	(15.0)	
Interest income	2.0	
Nonoperating income	4.0	
Earnings before taxes (EBT)	77.0	
Provision for income taxes	(18.0)	
Net income	59.0	

Step 3: Multiply the forecast ratio by next year's estimate of revenues (or appropriate forecast driver).

[1] Net PP&E = net property, plant, and equipment.

likely to change as you learn about the company, so at this point, a working model should be your priority. Once the entire model is complete, return to the forecast page and enter your best estimates.

3. *Multiply the forecast ratio by an estimate of its driver*. Since most line items are driven by revenues, most forecast ratios, such as cost of goods sold (COGS) to revenues, should be applied to estimates of future revenues. This is why a good revenue forecast is critical. Any error in the revenue forecast will be carried through the entire model. Ratios dependent on other drivers should be multiplied by their respective drivers.

Exhibit 13.4 presents the historical income statement and partially completed forecast for a hypothetical company. To demonstrate the three-step process, we forecast cost of goods sold. In the first step, calculate historical COGS as a function of revenues, which equals 37.5 percent. To start the model, initially set next year's ratio equal to 37.5 percent as well. Finally, multiply the forecast ratio by an estimate of next year's revenues: 37.5 percent × $288 million = $108 million.

Note that we did not forecast COGS by increasing the previous year's costs by 20 percent (the same growth rate as revenues). Although this process leads to the same *initial* answer, it reduces flexibility. By using a forecast ratio rather than a growth rate, we can either vary estimates of revenues (and COGS will change in step) or vary the forecast ratio (for instance, to value a potential improvement). If we had increased the COGS directly, however, we could only vary the COGS growth rate.

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EXHIBIT 13.5 **Typical Forecast Drivers for the Income Statement**

	Line item	Typical forecast driver	Typical forecast ratio
Operating	Cost of goods sold (COGS)	Revenue	COGS/revenue
	Selling, general, and administrative (SG&A)	Revenue	SG&A/revenue
	Depreciation	Prior-year net PP&E	Depreciation$_t$ / net PP&E$_{t-1}$
Nonoperating	Nonoperating income	Appropriate nonoperating asset, if any	Nonoperating income/nonoperating asset or growth in nonoperating income
	Interest expense	Prior-year total debt	Interest expense$_t$ / total debt$_{t-1}$
	Interest income	Prior-year excess cash	Interest income$_t$ / excess cash$_{t-1}$

Exhibit 13.5 presents typical forecast drivers and forecast ratios for the most common line items on financial statements. The appropriate choice for a forecast driver, however, depends on the company and the industry in which it competes.

Most valuation models, especially those of public companies, rely on ratios created directly from the company's financial statements. If you have access to other data that improves your forecast, incorporate it. For instance, the external valuation of a delivery company such as UPS will tie fuel costs directly to revenue. A more sophisticated model might tie fuel costs to the price of fuel and the number of packages delivered. Be mindful about incorporating new data, however. While additional data often improves the realism of your model, it will also increase its complexity. A talented modeler carefully balances realism with simplicity.

Operating Expenses For each operating expense on the income statement—such as cost of goods sold; selling, general, and administrative expenses; and research and development—we recommend generating forecasts based on revenues. In most cases, the process for operating expenses is straightforward. However, as outlined in Chapter 11, the income statement sometimes embeds certain nonoperating items in operating expenses. Before you begin the forecasting process, reformat the income statement to properly separate ongoing expenses from one-time charges.

Depreciation To forecast depreciation, you have three options. You can forecast depreciation as either a percentage of revenues or a percentage of property, plant, and equipment (PP&E), or—if you are working inside the company—you can also generate depreciation forecasts based on specific equipment purchases and depreciation schedules.

Although one can link depreciation to revenue, you will get better forecasts if you use PP&E as the forecast driver. To illustrate this, consider a company that makes a large capital expenditure every few years. Since depreciation is directly tied to a particular asset, it should increase only following an expenditure.

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If you tie depreciation to sales, it will incorrectly grow as revenues grow, even when capital expenditures haven't been made.

When using PP&E as the forecast driver, forecast depreciation as a percentage of net PP&E, rather than gross PP&E. Ideally, depreciation would be linked to gross PP&E, since depreciation for a given asset's life (assuming straight-line depreciation) equals gross PP&E divided by its expected life. But linking depreciation to gross PP&E requires modeling asset life and retiring the asset when it becomes fully depreciated. Implementing this correctly is tricky. If you forget to model asset retirements, for example, you would overestimate depreciation (and consequently its tax shield) in the later years.

If you have access to detailed, internal information about the company's assets, you can build formal depreciation tables. For each asset, project depreciation using an appropriate depreciation schedule, asset life, and salvage value. To determine company-wide depreciation, combine the annual depreciation of each asset.

Exhibit 13.6 presents a forecast of depreciation, as well as the remaining line items on the income statement.

Nonoperating Income Nonoperating income is generated by nonoperating assets, such as customer loans, nonconsolidated subsidiaries, and other equity investments. Since nonoperating income is typically excluded from free cash flow and the corresponding nonoperating asset is valued separately from core operations, the forecast will not affect the value of core operations. Instead, the primary purposes of nonoperating-income forecasts are cash flow planning and estimating earnings per share.

EXHIBIT 13.6 **Completed Forecast of the Income Statement**

Forecast worksheet			**Income statement**		
%	**2019**	**Forecast 2020**	$ million	**2019**	**Forecast 2020**
Revenue growth	20.0	20.0	Revenues	240.0	288.0
Cost of goods sold/revenues	37.5	37.5	Cost of goods sold	(90.0)	(108.0)
Selling and general expenses/revenues	18.8	18.8	Selling and general expenses	(45.0)	(54.0)
Depreciation$_t$/net PP&E$_{t-1}$	9.5	9.5	Depreciation	(19.0)	(23.8)
			EBITA	86.0	102.3
Interest rates			Interest expense	(15.0)	(13.8)
Interest expense	5.4	5.4	Interest income	2.0	1.2
Interest income	2.0	2.0	Nonoperating income	4.0	5.3
			Earnings before taxes (EBT)	77.0	95.0
Nonoperating items					
Nonoperating-income growth	33.3	33.3	Provision for income taxes	(18.0)	(22.2)
			Net income	59.0	72.7
Taxes					
Operating tax rate	23.4	23.4			
Statutory tax rate	24.0	24.0			
Effective tax rate	23.4	23.4			

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For nonconsolidated subsidiaries and other equity investments, the forecast methodology depends on how much information is available. For illiquid investments in which the parent company owns less than 20 percent, the company records income only when dividends are received or assets are sold at a gain or loss. For these investments, you cannot use traditional drivers to forecast cash flows; instead, estimate future nonoperating income by examining historical growth in nonoperating income or by examining the revenue and profit forecasts of publicly traded companies that are comparable to the equity investment.

For nonconsolidated subsidiaries with greater than 20 percent ownership, the parent company records income even when it is not paid out. Also, the recorded asset grows as the investment's retained earnings grow. Thus, you can estimate future income from the nonconsolidated investment either by forecasting a nonoperating-income growth rate or by forecasting a return on equity (nonoperating income as a percentage of the appropriate nonoperating asset) consistent with the industry dynamics and competitive position of the subsidiary.

Interest Expense and Interest Income Interest expense (or income) should be tied directly to the liability (or asset) that generates the expense (or income). The appropriate driver for interest expense is total debt. To simplify implementation, use *prior-year* debt to drive interest expense, rather than same year-end debt. To see why, consider a rise in operating costs. If the company uses debt to fund short-term needs, total debt will rise to cover the financing gap caused by lower profits. This increased debt load will cause interest expense to rise, dropping profits even further. The reduced level of profits, once again, requires more debt. To avoid the complexity of this feedback effect, compute interest expense as a function of the prior year's total debt. This shortcut will simplify the model and avoid circularity.[6]

A forecast of interest expense requires data from the income statement and the balance sheet. The balance sheet for our hypothetical company is presented in Exhibit 13.7. From the income statement presented in Exhibit 13.6, start with the 2019 interest expense of $15 million, and divide by 2018's total debt of $280 million (from the balance sheet, the sum of $200 million in short-term debt plus $80 million in long-term debt). This ratio equals 5.4 percent. To estimate the 2020 interest expense, multiply the estimated forecast ratio (5.35 percent) by 2019's total debt ($258 million), which leads to a forecast of $13.8 million. In this example, interest expense is falling even while revenues rise, because total debt is shrinking as the company generates cash from operations.

Using historical interest rates to forecast interest expense is a simple, straightforward estimation method. And since interest expense is not part of free cash flow, the choice of how to forecast interest expense will not affect the

[6] If you are using last year's debt multiplied by current interest rates to forecast interest expense, the forecast error will be greatest when year-to-year changes in debt are significant.

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EXHIBIT 13.7 **Historical Balance Sheet**

$ million

Assets	2018	2019	Liabilities and shareholders' equity	2018	2019
Operating cash	5.0	5.0	Accounts payable	15.0	20.0
Excess cash	100.0	60.0	Short-term debt	200.0	178.0
Inventory	35.0	45.0	Current liabilities	215.0	198.0
Current assets	140.0	110.0			
			Long-term debt	80.0	80.0
Net PP&E	200.0	250.0	Shareholders' equity	145.0	182.0
Equity investments	100.0	100.0	Total liabilities and equity	440.0	460.0
Total assets	440.0	460.0			

company's valuation (only free cash flow drives valuation; the cost of debt is modeled as part of the weighted average cost of capital).[7] When a company's financial structure is a critical part of the forecast, however, split debt into two categories: existing debt and new debt. Until repaid, existing debt should generate interest expense consistent with contractual rates reported in the company's financial notes. Interest expense based on new debt, in contrast, should be paid at current market rates, available from a financial data service. Projected interest expense should be calculated using a yield to maturity for comparably rated debt at a similar duration.

Estimate *interest income* the same way, with forecasts based on the asset generating the income. Be careful: interest income can be generated by multiple investments, including excess cash, short-term investments, customer loans, and other long-term investments. If a footnote details the historical relationship between interest income and the assets that generate the income (and the relationship is material), develop a separate calculation for each asset.

Income Taxes Do not forecast the provision for income taxes as a percentage of earnings before taxes. If you do, ROIC and FCF in forecast years will inadvertently change as leverage and nonoperating income change. Instead, start with a forecast of operating taxes on EBITA, and adjust for taxes related to nonoperating accounts, such as interest expense. Use this combined number to generate taxes on the income statement.

Exhibit 13.8 presents the forecast process for income taxes. To forecast operating taxes for 2020, multiply earnings before interest, taxes, and amortization (EBITA) by the *operating* tax rate (23.4 percent). Earlier, we estimated EBITA equal to $102.3 million for 2020. Do not use the statutory tax rate to forecast operating taxes. Many companies pay taxes at rates below their local statutory rate because

[7] In a WACC-based valuation model, the cost of debt and its associated tax shields are fully incorporated in the cost of capital. In an adjusted present value (APV) model, the interest tax shield is valued separately using a forecast of interest expense.

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EXHIBIT 13.8 **Forecast of Reported Taxes**

$ million

	2019	Forecast 2020
Operating taxes		
EBITA	86.0	102.3
× Operating tax rate	23.4%	23.4%
= Operating taxes	20.2	24.0
Taxes on nonoperating accounts		
Interest expense	(15.0)	(13.8)
Interest income	2.0	1.2
Nonoperating income	4.0	5.3
Nonoperating income (expenses), net	(9.0)	(7.3)
× Marginal tax rate	24.0%	24.0%
= Nonoperating taxes	(2.2)	(1.8)
Provision for income taxes[1]	18.0	22.2

[1] The provision for income taxes equals the sum of operating and nonoperating taxes.

of low foreign rates and operating tax credits.[8] Failure to recognize operating credits can cause errors in forecasts and an incorrect valuation. Also, if you use historical tax rates to forecast future tax rates, you implicitly assume that these special incentives will grow in line with EBITA. If this is not the case, EBITA should be taxed at the marginal rate, and tax credits should be forecast one by one.

Next, forecast the taxes related to nonoperating accounts. Although such taxes are not part of free cash flow, a robust forecast of them will provide insights about future net income and cash needs. For each line item between EBITA and earnings before taxes, compute the marginal taxes related to that item. If the company does not report each item's marginal tax rate, use the country's statutory rate. In Exhibit 13.8, the cumulative net nonoperating expense ($7.3 million in 2020) was multiplied by the marginal tax rate of 24 percent. It is possible to do this because each item's marginal income tax rate is the same. When marginal tax rates differ across nonoperating items, forecast nonoperating taxes line by line.

To determine the 2020 provision for income taxes, sum operating taxes ($24.0 million) and taxes related to nonoperating accounts (–$1.8 million). You now have a forecast of $22.2 million for reported taxes, calculated such that future values of FCF and ROIC will not change with leverage.

Step 4: Forecast the Balance Sheet: Invested Capital and Nonoperating Assets

To forecast the balance sheet, start with items related to invested capital and nonoperating assets. Do not forecast excess cash or sources of financing (such as debt and equity). Excess cash and sources of financing require special treatment and will be handled in step 5.

[8] For an in-depth discussion on the difference between statutory, effective, and operating tax rates, see Chapter 20.

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EXHIBIT 13.9 **Stock-versus-Flow Example**

	Year 1	Year 2	Year 3	Year 4
Revenues, $	1,000	1,100	1,200	1,300
Accounts receivable, $	100	105	121	120
Stock method				
Accounts receivable as a % of revenues	10.0	9.5	10.1	9.2
Flow method				
Change in accounts receivable as a % of change in revenues		5.0	16.0	(1.0)

When forecasting the balance sheet, one of the first issues you face is whether to forecast the line items in the balance sheet directly (in stocks) or indirectly by forecasting the year-to-year changes in accounts (in flows). For example, the stock approach forecasts end-of-year receivables as a function of revenues, while the flow approach forecasts the *change* in receivables as a function of the growth in revenues. We favor the stock approach. The relationship between the balance sheet accounts and revenues (or other volume measures) is more stable than that between balance sheet changes and changes in revenues. Consider the example presented in Exhibit 13.9. The ratio of accounts receivable to revenues remains within a tight band between 9.2 percent and 10.1 percent, while the ratio of changes in accounts receivable to changes in revenues ranges from –1 percent to 16 percent, too volatile to be insightful.

Exhibit 13.10 summarizes forecast drivers and forecast ratios for the most common line items on the balance sheet. The three primary operating line items are operating working capital, long-term capital such as net PP&E, and intangible

EXHIBIT 13.10 **Typical Forecast Drivers and Ratios for the Balance Sheet**

	Line item	Typical forecast driver	Typical forecast ratio
Operating line items	Operating working capital		
	Accounts receivable	Revenues	Accounts receivable/revenues
	Inventories	Cost of goods sold	Inventories/COGS
	Accounts payable	Cost of goods sold	Accounts payable/COGS
	Accrued expenses	Revenues	Accrued expenses/revenue
	Net PP&E	Revenues or units sold	Net PP&E/revenues
	Goodwill and acquired intangibles	Acquired revenues	Goodwill and acquired intangibles/acquired revenues
Nonoperating line items	Nonoperating assets	None	Growth in nonoperating assets
	Pension assets or liabilities	None	Trend toward zero
	Deferred taxes	Operating taxes or corresponding balance sheet item	Change in operating deferred taxes/operating taxes, or deferred taxes/corresponding balance sheet item

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EXHIBIT 13.11 **Partial Forecast of the Balance Sheet**

Forecast worksheet

Forecast ratio	2019	Forecast 2020
Working capital		
Operating cash, days' sales	7.6	7.6
Inventory, days' COGS	182.5	182.5
Accounts payable, days' sales	81.1	81.1
Fixed assets		
Net PP&E/revenues, %	104.2	104.2
Nonoperating assets		
Growth in equity investments, %	0.0	0.0

Balance sheet

$ million	2018	2019	Forecast 2020
Assets			
Operating cash	5.0	5.0	6.0
Excess cash	100.0	60.0	
Inventory	35.0	45.0	54.0
Current assets	140.0	110.0	
Net PP&E	200.0	250.0	300.0
Equity investments	100.0	100.0	100.0
Total assets	440.0	460.0	
Liabilities and equity			
Accounts payable	15.0	20.0	24.0
Short-term debt	200.0	178.0	
Current liabilities	215.0	198.0	
Long-term debt	80.0	80.0	
Shareholders' equity	145.0	182.0	
Total liabilities and equity	440.0	460.0	

assets related to acquisitions. Nonoperating line items include nonoperating assets, pensions, and deferred taxes, among others. We discuss each category next.

Operating Working Capital To start the balance sheet, forecast items within operating working capital, such as accounts receivable, inventories, accounts payable, and accrued expenses. Remember, operating working capital excludes any nonoperating assets (such as excess cash) and financing items (such as short-term debt and dividends payable).

When forecasting operating working capital, estimate most line items as a percentage of revenues or in days' sales.[9] Possible exceptions are inventories and accounts payable. Since these two accounts are economically tied to input prices, estimate them instead as a percentage of cost of goods sold (which is also tied to input prices).[10] Look for other links between the income statement and balance sheet that may exist. For instance, accrued wages can be calculated as a percent of compensation and benefits.

Exhibit 13.11 presents a partially completed forecast of our hypothetical company's balance sheet, in particular its operating working capital, long-term operating assets, and nonoperating assets (investor funds will be detailed later). All working-capital items are forecast in days, most of which are computed

[9] To compute a ratio in days' sales, multiply the percent-of-revenue ratio by 365. For instance, if accounts receivable equal 10 percent of revenues, this translates to accounts receivable at 36.5 days' sales. This implies that, on average, the company collects its receivables in 36.5 days.

[10] As a practical matter, we sometimes simplify the forecast model by projecting each working-capital item using revenues. The distinction is material only when price is expected to deviate significantly from cost per unit.

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using revenues. Working cash is estimated at 7.6 days' sales, inventory at 182.5 days' COGS, and accounts payable at 81.1 days' COGS. We forecast in days for the added benefit of tying forecasts more closely to the velocity of operating activities. For instance, if management announces its intention to reduce its inventory holding period from 180 days to 120 days, it is possible to compute changes in value by adjusting the forecast directly.

Property, Plant, and Equipment Consistent with our earlier argument concerning stocks and flows, net PP&E should be forecast as a percentage of revenues.[11] A common alternative is to forecast capital expenditures as a percentage of revenues. However, this method too easily leads to unintended increases or decreases in capital turnover (the ratio of PP&E to revenues). Over long periods, companies' ratios of net PP&E to revenues tend to be quite stable, so we favor the following three-step approach for PP&E:

1. Forecast net PP&E as a percentage of revenues.
2. Forecast depreciation, typically as a percentage of gross or net PP&E.
3. Calculate capital expenditures by summing the projected increase in net PP&E plus depreciation.

To continue our example, we use the forecasts presented in Exhibit 13.11 to estimate expected capital expenditures. In 2019, net PP&E equaled 104.2 percent of revenues. If this ratio is held constant for 2020, the forecast of net PP&E equals $300 million. To estimate capital expenditures, compute the increase in net PP&E from 2019 to 2020, and add 2020 depreciation from Exhibit 13.6.

$$\begin{aligned}\text{Capital Expenditures} &= \text{Net PP\&E}_{2020} - \text{Net PP\&E}_{2019} + \text{Depreciation}_{2020}\\ &= \$300.0 \text{ million} - \$250.0 \text{ million} + \$23.8 \text{ million}\\ &= \$73.8 \text{ million}\end{aligned}$$

For companies with low growth rates and projected improvements in capital efficiency, this methodology may lead to negative capital expenditures (implying asset sales). Although positive cash flows generated by equipment sales are possible, they are unlikely. In these cases, make sure to assess the resulting cash flow carefully.

Goodwill and Acquired Intangibles A company records goodwill and acquired intangibles when the price it pays for an acquisition exceeds the target's book value.[12] For most companies, we choose not to model potential

[11] Some companies, such as oil refiners, will report number of units. In these cases, consider using number of units instead of revenue to forecast equipment purchases.

[12] This section refers to acquired intangibles only. Forecast internal investments in intangibles, such as capitalized software and purchased sales contracts, with the methodology used for capital expenditures and PP&E.

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acquisitions explicitly, so we set revenue growth from new acquisitions equal to zero and hold goodwill and acquired intangibles constant at their current level. We prefer this approach because of the empirical literature documenting how the typical acquisition fails to create value (any synergies are transferred to the target through high premiums). Since adding a zero-NPV investment will not increase the company's value, forecasting acquisitions is unnecessary. In fact, by forecasting acquired growth in combination with the company's current financial results, you make implicit (and often hidden) assumptions about the present value of acquisitions. For instance, if the forecast ratio of goodwill to acquired revenues implies positive NPV for acquired growth, increasing the growth rate from acquired revenues can dramatically increase the resulting valuation, even when good deals are hard to find.

If you decide to forecast acquisitions, first assess what proportion of future revenue growth they are likely to provide. For example, consider a company that generates $100 million in revenues and has announced an intention to grow by 10 percent annually—5 percent organically and 5 percent through acquisitions. In this case, measure historical ratios of goodwill and acquired intangibles to acquired revenues, and apply those ratios to acquired revenues. For instance, assume the company historically adds $3 in goodwill and intangibles for every $1 of acquired revenues. Multiplying the expected $5 million of acquired growth by 3, you obtain an expected increase of $15 million in goodwill and acquired intangibles. Make sure, however, to perform a reality check on your results by varying acquired growth and observing the resulting changes in company value. Confirm that your results are consistent with the company's past performance related to acquisitions and the challenges of creating value through acquisition.

Nonoperating Assets, Unfunded Pensions, and Deferred Taxes Next, forecast nonoperating assets (such as nonconsolidated subsidiaries), debt equivalents (such as pension liabilities), and equity equivalents (such as deferred taxes). Because many nonoperating items are valued using methods other than discounted cash flow (see Chapter 16), any forecasts of these items are primarily for the purpose of financial planning and cash management, not enterprise valuation. For instance, consider unfunded pension liabilities. Assume management announces its intention to reduce unfunded pensions by 50 percent over the next five years. To value unfunded pensions, do not discount the projected outflows over the next five years. Instead, use the current actuarial assessments of the shortfall, which appear in the note on pensions. The rate of reduction will have no valuation implications but will affect the ability to pay dividends or may require additional financing. To this end, model a reasonable time frame for eliminating pension shortfalls.

We are extremely cautious about forecasting (and valuing) nonconsolidated subsidiaries and other equity investments. Valuations should be based

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on assessing the investments currently owned, not on discounting the forecast changes in their book values and/or their corresponding income. If a forecast is necessary for planning, keep in mind that income from associates is often noncash, and nonoperating assets often grow in a lumpy fashion unrelated to a company's revenues. To forecast equity investments, rely on historical precedent to determine the appropriate level of growth.

Regarding deferred-tax assets and liabilities, those used to occur primarily through differences in depreciation schedules (investor and tax authorities use different depreciation schedules to determine taxable income). Today, deferred taxes arise for many reasons, including tax adjustments for pensions, stock-based compensation, acquired-intangibles amortization, and deferred revenues (see Chapter 20 for an in-depth discussion of deferred taxes).

For sophisticated valuations that require extremely detailed forecasts, forecast deferred taxes line by line, tying each tax to its appropriate driver. In most situations, forecasting operating deferred taxes by computing the aggregate proportion of taxes likely to be deferred will lead to reasonable results. For instance, if operating taxes are estimated at 23.4 percent of EBITA and the company historically could defer one-fifth of operating taxes paid, we often assume it can defer one-fifth of 23.4 percent going forward. Operating-related deferred-tax liabilities will then increase by the amount deferred.

Step 5: Reconcile the Balance Sheet with Investor Funds

To complete the balance sheet, forecast the company's sources of financing. To do this, rely on the rules of accounting. First, use the principle of clean surplus accounting:

$$\begin{aligned}\text{Equity}_{2020} &= \text{Equity}_{2019} + \text{Net Income}_{2020} \\ &\quad - \text{Dividends}_{2020} + \text{Net Equity Issued}_{2020}\end{aligned}$$

Applying this to our earlier example, Exhibit 13.12 presents the statement of shareholders' equity. To estimate equity in 2020, start with 2019 equity of $182 million from Exhibit 13.11. To this value, add the 2020 forecast

EXHIBIT 13.12 **Statement of Shareholders' Equity**

$ million

	2018	2019	Forecast 2020
Shareholders' equity, beginning of year	120.8	145.0	182.0
Net income	40.2	59.0	72.7
Dividends	(16.0)	(22.0)	(27.1)
Issuance (repurchase) of common stock	–	–	–
Shareholders' equity, end of year	145.0	182.0	227.6
Dividends/net income, %	39.8	37.3	37.3

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EXHIBIT 13.13 **Forecast Balance Sheet: Sources of Financing**

$ million

	2018	2019	Preliminary 2020F	Completed 2020F
Assets				
Operating cash	5.0	5.0	6.0	6.0
Excess cash	100.0	60.0		49.6
Inventory	35.0	45.0	54.0	54.0
Current assets	140.0	110.0	60.0	109.6
Net PP&E	200.0	250.0	300.0	300.0
Equity investments	100.0	100.0	100.0	100.0
Total assets	440.0	460.0	460.0	509.6
Liabilities and equity				
Accounts payable	15.0	20.0	24.0	24.0
Short-term debt	200.0	178.0	178.0	178.0
Current liabilities	215.0	198.0	202.0	202.0
Long-term debt	80.0	80.0	80.0	80.0
Newly issued debt	–	–		–
Shareholders' equity	145.0	182.0	227.6	227.6
Total liabilities and equity	440.0	460.0	509.6	509.6

Step 1: Determine retained earnings using the clean surplus relation, forecast existing debt using contractual terms, and keep common stock constant.

Step 2: Test which is higher: (a) assets excluding excess cash or (b) liabilities and equity, excluding newly issued debt.

Step 3: If assets excluding excess cash are higher, set excess cash equal to zero, and plug the difference with the newly issued debt. Otherwise, plug with excess cash.

of net income: $72.7 million from the income statement in Exhibit 13.6. Next, estimate the dividend payout. In 2019, the company paid out 37.3 percent of net income in the form of dividends. Applying a 37.3 percent payout ratio to estimated net income leads to $27.1 million in expected dividends. Finally, add new equity issued net of equity repurchased, which in this example is zero. Using the clean surplus relationship, we estimate 2020 equity at $227.6 million.

At this point, four line items on the balance sheet remain: excess cash, short-term debt, long-term debt, and a new account titled "newly issued debt." Some combination of these line items must make the balance sheet balance. For this reason, these items are often referred to as "the plug." In simple models, existing debt either remains constant or is retired on schedule, according to contractual terms.[13] To complete the balance sheet, set one of the remaining two items (excess cash or newly issued debt) equal to zero. Then use the primary accounting identity—assets equal liabilities plus shareholders' equity—to determine the remaining item.

Exhibit 13.13 presents the elements of this process for our example. First, hold short-term debt, long-term debt, and common stock constant. Next, sum total assets, excluding excess cash: cash ($6 million), inventory ($54 million), net PP&E ($300 million), and equity investments ($100 million) total $460 million. Then sum total liabilities and equity, excluding newly

[13] Given the importance of debt in a leveraged buyout, buyout models often contain a separate worksheet detailing interest and principal repayment by year for each debt contract.

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issued debt: accounts payable ($24 million), short-term debt ($178 million), long-term debt ($80 million), and shareholders' equity ($227.6 million) total $509.6 million. Because liabilities and equity (excluding newly issued debt) are greater than assets (excluding excess cash), newly issued debt is set to zero. Now total liabilities and equity equal $509.6 million. To ensure that the balance sheet balances, we set the only remaining item, excess cash, equal to $49.6 million. This increases total assets to $509.6 million, and the balance sheet is complete.

To implement this procedure in a spreadsheet, use the spreadsheet's prebuilt If function. Set up the function so it sets excess cash to zero when assets (excluding excess cash) exceed liabilities and equity (excluding newly issued debt). Conversely, if assets are less than liabilities and equity, the function should set short-term debt equal to zero and excess cash equal to the difference.

The Link Between Capital Structure Forecasts and Valuation When using excess cash and newly issued debt to complete the balance sheet, you will likely encounter one common side effect: as growth drops, newly issued debt will drop to zero, and excess cash will become very large.[14] But what if a drop in leverage is inconsistent with your long-term assessments concerning capital structure? In an enterprise DCF valuation that uses the weighted average cost of capital for discounting, this side effect does not matter. Excess cash and debt are not included as part of free cash flow, so they do not affect the enterprise valuation. Capital structure affects enterprise DCF only through the weighted average cost of capital.[15] Thus, only an adjustment to WACC will lead to a change in valuation.

To bring the capital structure on the balance sheet in line with the capital structure implied by WACC, adjust the dividend payout ratio or amount of net share repurchases. For instance, as the dividend payout is increased, retained earnings will drop, and this should cause excess cash to drop as well. By varying the payout ratio (both dividends and share repurchases), you can also test how robust your FCF model is. Specifically, ROIC and FCF, and hence value, should not change when the dividend rate or amount of share repurchases is adjusted.

How you choose to model the payout ratio depends on the requirements of the model. In most situations, you can adjust the dividend payout ratio

[14] Whenever ROIC is greater than revenue growth, a company will generate operating cash flow; that is, the investment rate will be negative. If dividends or share repurchases are not increased to disgorge cash, debt will drop, and/or excess cash will accumulate.

[15] In the APV model, your forecast of debt will affect valuation. Interest tax shields are computed year by year based on the amount of debt, the interest rate, and the tax rate. Models that discount with a constant WACC implicitly assume debt-to-value never changes, such that balance sheet forecasts are ignored.

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or amount of repurchases by hand when needed (remember, the ratio does not affect value but rather brings excess cash and newly issued debt closer to reality). For more complex models, determine net debt (total debt less excess cash) by applying the target net-debt-to-value ratio modeled in the WACC at each point in time. Next, using the target debt-to-value ratio, solve for the required payout. To do this, however, you must perform a valuation in each forecast year and iterate backward—a time-consuming process for a feature that will not affect the final valuation.[16]

Step 6: Calculate ROIC and FCF

Once you have completed your income statement and balance sheet forecasts, calculate ROIC and FCF for each forecast year. This process should be straightforward if you have already computed ROIC and FCF historically. Since a full set of forecast financials is now available, merely copy the two calculations from historical financials to projected financials.

For companies that are creating value, future ROICs should fit one of three general patterns: ROIC should either remain near current levels (when the company has a distinguishable sustainable advantage), trend toward an industry or economic median, or trend to the cost of capital. Think through the economics of the business to decide what is appropriate. For more on long-term trends of ROIC, refer to Chapter 8.

ADVANCED FORECASTING

The preceding sections detailed the process for creating a comprehensive set of financial forecasts. When forecasting, you are likely to come across three advanced issues: forecasting using nonfinancial operating drivers, forecasting using fixed and variable costs, and handling the impact of inflation.

Nonfinancial Operating Drivers

Until now, the chapter has created forecasts that rely solely on financial drivers. In industries where prices are changing or technology is advancing, forecasts should incorporate nonfinancial ratios, such as volume and productivity.

Consider the turmoil in the airline industry during the early 2000s. Fares requiring Saturday-night stays and advance purchases disappeared as

[16] To value Costco in Appendix H, we modeled a constant leverage ratio year by year and iterated backward. While iteration is not necessary to value a company more generally, it is required to ensure that the enterprise DCF valuation ties to other valuation methodologies, such as cash-flow-to-equity models.

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competition from low-cost carriers intensified. Network carriers could no longer distinguish business travelers, their primary source of profit, from leisure travelers. As the average price dropped, costs rose as a percentage of sales. But were airlines truly becoming higher-cost?[17] And how would this trend continue? To forecast changes more accurately, it is necessary to separate price from volume (as measured by seat-miles). Then, instead of forecasting costs as a percentage of revenues, forecast costs as a function of expected quantity—in this case, seat-miles.

The same concept applies to advances in technology. For instance, rather than estimate labor as a percentage of revenues, you could forecast units per employee and average salary per employee. Separating these two drivers of labor costs allows you to model a direct relationship between productivity improvements from new technology and estimated changes in units per employee.

Fixed versus Variable Costs

When you are valuing a small project, it is important to distinguish fixed costs (incurred once to create a basic infrastructure) from variable costs (correlated with volume). When you are valuing an individual project, only variable costs should be increased as revenues grow.

At the scale of most publicly traded companies, however, the distinction between fixed and variable costs is often immaterial, because nearly every cost is variable. For instance, consider a mobile-phone company that transmits calls using radio-frequency towers. In spite of the common perception that the tower is a fixed cost, this is true for only a given number of subscribers. As subscribers increase beyond a certain limit, new towers must be added, even in an area with preexisting coverage. (A small company adding 1,000 customers can leverage economies of scale more than a large company adding 100,000 customers.) What is a fixed cost in the short run for small increases in activity becomes variable over the long run even at reasonable growth rates (10 percent annual growth doubles the size of a company in about seven years). Since corporate valuation is about long-run profitability and growth, nearly every cost should be treated as variable.

When an asset, such as computer software or a mobile app, is truly scalable, its development cost should be treated as a fixed cost. Be careful, however. Many technologies, such as computer software, quickly become obsolete, requiring new incremental expenditures for the company to remain competitive. In this case, a cost deemed fixed actually requires repeated cash outflows, just not in traditional ways.

[17] For example, Spirit Airlines dedicates a higher percentage of revenue to labor than American Airlines does. In terms of cost per seat-mile, however, American is the higher-cost airline of the two.

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Incorporating Inflation

In Chapter 10, we recommended that financial-statement forecasts and the cost of capital be estimated in nominal currency units (with inflation), rather than real currency units (without inflation). To remain consistent, the nominally based financial forecast and the nominally based cost of capital must reflect the same expected general inflation rate. This means the inflation rate built into the forecast must be derived from an inflation rate implicit in the cost of capital.[18]

When possible, derive the expected inflation rate from the term structure of government bond rates. The nominal interest rate on government bonds reflects investor demand for a real return plus a premium for expected inflation. Estimate expected inflation as the nominal rate of interest less an estimate of the real rate of interest, using the following formula:

$$\text{Expected Inflation} = \frac{(1 + \text{Nominal Rate})}{(1 + \text{Real Rate})} - 1$$

To estimate expected inflation, start by calculating the nominal yield to maturity on a ten-year government bond. But how do you find the real rate? Many countries, such as the United States, United Kingdom, and Japan, issue inflation-linked bonds (ILBs). An ILB is a bond that protects against inflation by growing the bond's coupons and principal at the consumer price index (CPI). Consequently, the yield to maturity on an ILB is the market's expectation of the real interest rate over the life of the bond. In March 2019, the yield on a ten-year U.S. Treasury bond equaled 2.57 percent, and the yield on a U.S. Treasury inflation-protected security (TIPS) bond equaled 0.66 percent.[19] Unlike previous decades, when the real rate hovered around 2 percent, the real rate has been volatile during the last ten years, even dropping below zero in 2012. To determine expected inflation, apply the previous formula to the data:

$$\text{Expected Inflation} = \frac{1.0257}{1.0066} - 1 = 0.0190$$

Expected inflation, as measured by the difference in nominal and real bonds, equals 1.90 percent annually over the next ten years.

[18] Individual line items may have inflation rates that are higher or lower than the general rate, but they should still derive from the general rate. For example, the revenue forecast should reflect the growth in units sold and the expected increase in unit prices. The increase in unit prices, in turn, should reflect the generally expected level of inflation in the economy plus or minus an inflation rate differential for that specific industry. Suppose general inflation is expected to be 4 percent and unit prices for the company's products are expected to increase at one percentage point less than general inflation. Overall, the company's prices would be expected to increase at 3 percent per year. If we assume a 3 percent annual increase in units sold, we would forecast 6.1 percent annual revenue growth (1.03 × 1.03 – 1).

[19] 10-Year Treasury Constant Maturity Rate (DGS10) and 10-Year Treasury Inflation-Indexed Security, Constant Maturity (FII10), Federal Reserve Bank of St. Louis.

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EXHIBIT 13.14 **Expected Inflation versus Growth in the Consumer Price Index**

Source: Federal Reseve Bank of St. Louis.

Exhibit 13.14 presents annualized growth in the U.S. consumer price index (CPI) versus expected ten-year inflation implied by traditional U.S. Treasury bonds and U.S. TIPS bonds. Since the ten-year TIPS bond is based on long-term inflation, the implied inflation rate is much more stable than the one-year change in CPI (in mid-2008, CPI grew at more than 5 percent when crude oil spiked, only to crater after the recession as companies cut prices to generate demand). Since 2000, actual and implied inflation have both hovered around 2 percent annually.

Inflation can distort historical analysis, especially when it exceeds 5 percent annually. In these situations, historical financials should be adjusted to reflect operating performance independent of inflation. We discuss the impact of high inflation rates in Chapter 26.

CONCLUDING THOUGHTS

In this chapter, we provided a detailed line-by-line process to create a set of financial forecasts. While it is important that the model reflect the complexities of the business you are analyzing, always keep a close eye on the bigger picture. Make sure resulting value drivers, such as ROIC and growth, are consistent with the past performance of the business and the industry's economics. When the model is complete, use the model to test the importance of various inputs. A sensitivity table can provide insight on not only the valuation but also on the actions management must undertake to capture it.

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REVIEW QUESTIONS

1. HouseholdCo, a large household products firm, recently reported $6.2 billion in "Accrued and other liabilities" and $49.0 billion in revenues, such that accrued and other liabilities equaled 12.6 percent of revenues. Using data from the annual report's Footnote 4, provided in Exhibit 13.15, discuss why a forecast ratio of 12.6 percent going forward would distort your forecast of free cash flow. How should the balance sheet be reorganized to prevent this?

EXHIBIT 13.15 **HouseholdCo: Annual Report**

Note 4
Supplemental financial information
Selected components of current and noncurrent liabilities were as follows:

Accrued and other liabilities—current
$ million

	Year 1
Marketing and promotion	2,833.0
Compensation expenses	1,846.0
Accrued ShaveCo exit costs*	456.0
Taxes payable	922.0
Other	102.0
Total	6,159.0

*Results from severance costs due to acquisition of ShaveCo, a consumer products firm, in prior year.

2. Exhibit 13.16 presents the income statement and balance sheet for PartsCo, a $1.2 billion supplier of machinery parts. The company is expected to increase revenues by 8 percent annually for the next five years. Using the methodology outlined in Exhibit 13.4 and the template in Exhibit 13.16, forecast the next five years of income statements for PartsCo. Assume that the next five years' forecast ratios are identical to this year's ratios. Forecast depreciation as a percentage of the prior year's property, plant, and equipment. Forecast interest as a percentage of the prior year's total debt.

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Exhibit 13.16 **PartsCo: Consolidated Financial Statements**

$ million

Income statement	Prior year	Current year	Forecast year 1	Forecast year 2	Forecast year 3	Forecast year 4	Forecast year 5
Revenues	1,100.0	1,188.0					
Cost of sales	(825.0)	(891.0)					
Selling costs	(165.0)	(178.2)					
Depreciation	(27.6)	(29.7)					
Operating profit	82.4	89.1					
Interest expense	(15.0)	(15.0)					
Earnings before taxes	67.4	74.1					
Taxes	(20.2)	(22.2)					
Net income	47.2	51.9					
Dividends	19.0	21.0					

Balance sheet	Prior year	Current year	Forecast year 1	Forecast year 2	Forecast year 3	Forecast year 4	Forecast year 5
Assets							
Working cash	22.0	23.8					
Excess cash and marketable securities	385.2	377.6					
Accounts receivable	110.0	118.8					
Inventory	123.8	133.7					
Current assets	641.0	653.9					
Property, plant, and equipment	440.0	475.2					
Equity investments	320.0	320.0					
Total assets	1,401.0	1,449.1					
Liabilities and equity							
Accounts payable	123.8	133.7					
Short-term debt	90.0	90.0					
Accrued expenses	88.0	95.0					
Current liabilities	301.8	318.7					
Long-term debt	210.0	210.0					
Newly issued debt	-	-					
Shareholders' equity	889.2	920.4					
Liabilities and equity	1,401.0	1,449.1					

3. Using the methodology outlined in Exhibit 13.11, forecast the operating items on the next five years of balance sheets for PartsCo. Forecast each balance sheet item as a function of revenues, except inventory and accounts payable, which should be forecast as a function of cost of sales. Your forecast should be consistent with the revenues and cost of sales forecast in Question 2. (Note that Exhibit 13.11 converts figures to number of days. You do not need to go this extra step. You can just compute the historical ratios as a percentage of either revenues or cost of goods sold and use those ratios for the forecast.)

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4. Using the methodology outlined in Exhibits 13.12 and 13.13, forecast the financing items on next year's balance sheet for PartsCo. Assume that short-term debt remains at $90 million, long-term debt remains at $210 million, no equity is raised, and the firm maintains the same dividend payout ratio as the current year. Use either newly issued debt or excess cash and marketable securities as the funding plug to balance the balance sheet. Your forecast should be consistent with the forecasts in Questions 2 and 3.
5. The chief financial officer of PartsCo has asked you to rerun the forecast of the company's income statement and balance sheet at a growth rate of 3 percent. If the company generates more cash than it needs, how could the balance sheet be adjusted to handle this? What alternatives exist to handle new cash?
6. The chief financial officer of PartsCo returns again and explains that there is a possibility of a huge contract being awarded to the company. If this occurs, then the company is expected to achieve 25 percent revenue growth annually for the next five years. How does this affect the pro forma balance sheet? Which accounts are notably different than in the prior situations?

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